COMPUTER ARCHITECTURE AND ORGANISATION

Computer Hardware and Software

INTRODUCTION

Understanding computer hardware is crucial because it forms the foundation of all computing processes. It helps improve system performance, troubleshoot technical issues, and make cost-effective decisions when purchasing or upgrading hardware. Key components such as input devices (like keyboards and sensors), output devices (like monitors and printers), and storage devices (like hard drives and SSDs) play essential roles in how computers interact with users and the world. Additionally, hardware knowledge connects the digital and physical worlds, enabling the creation and optimisation of real-world applications in areas like robotics, healthcare, and automation. An understanding of how hardware components, including input, output, and storage devices function, allows students to better understand how software interacts with these physical systems, enhancing their overall understanding of computing.

Consider how computer hardware technologies—such as input devices (like keyboards, touchscreens, and microphones), output devices (like monitors, speakers, and printers), and storage devices (like hard drives and cloud services)—are integral parts of your everyday life. Think about how you use these components daily when you text on your smartphone, listen to music, watch videos, or save files for school projects. By reflecting on how these technologies are embedded in your routine, you can appreciate the importance of understanding hardware and its role in connecting the digital and physical aspects of your world.

Hardware components of a computer are various physical parts or devices that work together to provide the necessary functionality for computing tasks.

Hardware devices are categorised into the following:

- Input devices
- Output devices
- Processing devices
- Storage devices
- Communication devices

We will now look at each of these in more detail, covering examples of each category.

At the end of this section, you should be able to:

- Explain the features of the hardware components of a computer (input hardware, processing hardware, output hardware, storage hardware and communication hardware).
- Describe the categories of Computer Software

Key Ideas

- Hardware is the physical part of the computer.
- Input hardware is devices used to enter data and instructions into the computer.
- Output hardware displays the data/information.
- Storage hardware keeps the data for future use.
- Communication hardware sends and receives data and signals.
- Network refers to a collection of interconnected devices that communicate with each other to share resources, data, and services.
- Network storage is a type of hardware that allows access to storage on a network.
- Cloud storage is a type of storage that allows access to storage in remote servers in data centres through the Internet.
- Communication hardware facilitates data transfer between systems or devices in a network.
- The motherboard is a circuit board of the computer. RAM slots, expansion slots, and CPU sockets are some of the components of the CPU.
- A software is a set of instructions that tells a computer what to do. It includes a broad range of applications or programs, tools, and systems that enable the hardware to perform specific tasks.
- A computer program is a specific set of instructions that a computer executes to perform a particular task. A program is a subset of software, usually focused on achieving a single function.

INPUT DEVICES

Input devices are hardware components that allow users to give data and instructions to a computer system. They enable users to interact with the computer and provide the necessary input for executing tasks. Here are some examples of input devices:

1. Keyboards

A keyboard is a common input device, consisting of a set of keys that allow users to input alphanumeric characters, symbols, and commands. They facilitate the typing text and providing commands to the computer.



Figure 2.1 Keyboard.

2. Mouse

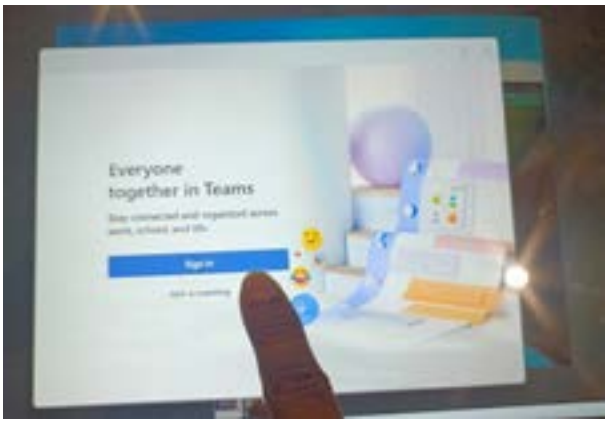
The mouse is a pointing device that allows users to control the cursor on the screen. They typically have buttons and a scroll wheel, enabling users to select objects, navigate interfaces, and perform actions through mouse movements and clicks. It is important to know that the plural form of mouse is “mice”.



Figure 2.2 Mouse.

3. Touchscreens

Touchscreens enable users to directly interact with the computer by touching the display. They can detect and respond to finger gestures, allowing for intuitive input methods such as tapping, swiping, or pinching. Playing a similar role to the mouse, they are particularly popular for portable deceives where a mouse would be impractical.



4. Scanners

Scanners are used to convert physical documents, images, or objects into digital formats. They capture the data optically and transfer it to the computer for storage, editing, or further processing.



Figure 2.3 Scanners.

Activity 2.1

1. A computer lab of Mpeasem SHS located at Kojokrom in the Western Region of Ghana needs to accommodate students with diverse needs, including those with disabilities and different backgrounds. As a computing student studying input devices, suggest a plan to equip the lab with a variety of input devices, ensuring accessibility for all students in groups. You might want to research more input devices, perhaps those with accessibility in mind. Discuss your group reports with your teacher.
2. Individually move around the computer room and write down the names of input devices you can see.

OUTPUT DEVICES

The output device displays the result of the processed data that is entered into the computer through an input device. There are a few output devices that display output in different ways such as text, images, hard copies, and audio or video.

They bridge the gap between digital data and human perception, letting users engage with computer-generated information.

There are many types of output devices for processed data as text, images, or video.

Monitors and projectors allow users to view computer output on screens or project it onto larger surfaces.

1. Monitors

Monitors allow users to view computer output on screens. There are different types of monitors, three of which as shown in Figure 2.4.



Figure 2.4 Monitors.

Some computers, such as laptops, have built-in monitors.

Monitors can be found where computers are being used. For example, they are used in educational institutions, such as schools and universities, to facilitate learning. They display educational content, multimedia presentations, interactive learning materials, etc.

2. Projectors

A projector is an output device that enables the user to project the output onto a big screen or wall. It can be connected to a computer and similar devices to project their output onto a screen. It uses light and lenses to produce magnified texts, images, and videos. So, it is an ideal output device to give presentations or to teach a large number of people.

Modern projects (digital projectors) come with multiple input sources such as Display Port (DP) and High-Definition Multimedia Interface (HDMI) ports for newer equipment and Video Graphics Array (VGA) ports that support older devices. Some projectors are designed to support

Wi-Fi and Bluetooth as well. They can be fixed onto the ceiling, placed on a stand, and more and are frequently used for classroom teaching, giving presentations, home cinemas, etc.



Figure 2.5 A projector.

3. Plotters

A plotter is a specialised output device used to generate high-quality, accurate, and detailed graphics. It's popular in fields like engineering, architecture, and

graphic design. Unlike printers, which use ink or toner to create pictures or text on paper, plotters can utilise a pen or marker to draw continuous lines on diverse media such as paper, vinyl, or film.

Examples of Plotters

Pen Plotters: Pen plotters are the most classic sort of plotter, drawing continuous lines on paper or other media using a pen or marker.



Figure 2.6 Pen Plotter.

Electrostatic Plotters: Electrostatic plotters use an electrostatic charge to attract toner or ink onto paper. They operate by selectively charging areas of the paper and then applying toner or ink to those charged areas, resulting in the formation of graphical output.



Figure 2.7 Electrostatic Plotter.

4. Printers

A printer produces hard copies/physical copies of the processed data. It enables the user to print images, text or any other information onto the paper. Users can pick from various printer types, such as inkjet and laser, to meet their printing demands. Printers enable the generation of tangible copies for record-keeping, presentations, marketing materials, and other purposes, from household to professional settings. Printers are essential for personal and commercial use due to their simplicity and adaptability.



Figure 2.8 Small Laser Printer.

Printers vary in type. size, speed, and cost. Some printers can output in A3. Some printers have photocopy and scan functions. Popular types of printers are laser printers and inkjet printers.

3D printers (as shown in Figure 2.9) create three-dimensional objects by depositing layers of material (plastic, metal, or composite) based on a digital model. They are used in various industries, including manufacturing, prototyping, and healthcare.



Figure 2.9 3D printing.

5. Sound Output devices

These devices deliver audio output for listening or communication purposes. Speakers and headphones/earphones enable users to hear the sound - music, and speech.



Figure 2.10 Examples of Sound Output Devices.

Activity 2.2

1. Abdulai Joanna Fati hailed from the Northern Region of Ghana and works as an Internet Café Manageress of DM ICT Centre at Ho. She wants to change the old printers of the café. As a hardware specialist who studied output devices, recommend describing to her at least three (3) things that she should consider when buying a new printer.
2. Individually move around the computer room and write down the name of the devices there and their categories (input or output).

3. Upon your admission, describe how the admission officer used the various input and output devices to help you get the hard copy of your admission letter, prospectus and code of conduct online.

STORAGE DEVICES

These devices are essential components of a computer system that store and retrieve data, programs, and files. They provide long-term data storage and enable users to save, access, and manage their digital information.

When looking at storage devices, there are three considerations: capacity, speed, and cost (how much to purchase per MB).

Storage Capacity & Performance

a. One characteristic of storage devices is their capacity, which is the amount of data they can store, typically measured in Kilobytes (KB), Megabytes (MB), Gigabytes (GB), or Terabytes (TB).

b. Performance factors include read/write speeds, access times and data transfer rates.

Types of storage devices

Storage devices fall into three categories: magnetic, optical and flash.

1. Magnetic

Magnetic storage uses a magnetisable material. Patterns of magnetisation are then used to represent binary sequences. Magnetic storage tends to have a high capacity at low cost. One particular downside to Magnetic storage is the lower read speeds, typically they are relatively slow. Examples include magnetic tape and hard disk drives.

Magnetic tape is sometimes used for backup of server computers.

Hard Disk Drives (HDDs) are the most common form of magnetic storage devices used in computers. They use magnetic storage to store data on spinning disks (platters) and read/write heads to access and modify data. HDD capacities typically range from several hundred gigabytes (GB) to several terabytes (TB).



Figure 2 .11 Internal HDDs.

2. Optical

Optical storage works by using laser technology to read and write data a laser. Examining the reflection from the laser will determine whether there are pits on the surface of the disk representing 1s, or lands representing 0s. Examples of optical disks are CDs, DVDs. They can be used for data storage, software installation, media playback and creating backups.

Optical disk drives are gradually being phased out in favour of faster and more versatile storage options.



Figure 2.12 Optical Disks (DVDs).

3. Flash

Flash memory is a type of non-volatile solid-state memory that stores data by trapping electrons in tiny cells to represent ones and zeros. It retains information even when the power is off, allowing data to be stored long-term without needing electricity. Because it has no moving parts and can be quickly read and written, flash memory is used in devices like USB flash drives, SSDs, and memory cards.

Solid State Drives (SSDs)

SDDs offer faster data access and transfer speeds compared to HDDs due to the absence of moving parts. SSDs are more resistant to physical shocks and consume less power compared to HDDs. They come in various form factors and can be internal or external/portable.



Figure 2.13 Internal Solid-State Drive (SSD).

USB Flash Drives

USB flash drives, also known as thumb drives, pen drives or USB sticks, are portable storage devices that use flash memory. They connect to computers via USB ports and offer a convenient way to transfer and store data. They come in various storage capacities and are widely used for data ***backup, file transfer, and portable storage.***



Figure 2.14 Flash Drives. *Source:* Google images

Memory Cards

Memory cards are small, portable storage media commonly used in cameras, smartphones, tablets, and other portable devices. The data can usually be read by connecting the device with the card to the computer or removing the card from the device and inserting it into a memory card reader connected to the computer.

They provide removable and expandable storage options and use flash memory technology.

Popular memory card formats include Secure Digital (SD) cards, microSD cards, Compact Flash (CF) cards, and Memory Stick.



Figure 2.15 SD Cards

Activity 2.3

1. Use the internet to find out about the following pair of devices/ media: HDDs vs SDDs, HDDs vs Magnetic Tape Drives, CDs vs DVDs, USB flash drive vs memory card. Compare these devices/media and use a table in Word to present your work.
2. Complete the following table template (column headings: device name, typical use(s), main features, average cost, image) and then populate as much as possible with no online assistance. Report your result to your teacher.

S/N	Device Name	Typical Use(S)	Main Features	Average Cost	Image
1					
2					
3					
4					
5					
6					
7					
8					
9					
10					

3. In groups, observe the picture of hardware devices in Figure 2.16 and answer the questions that follow:



Figure 2.16 Hardware Devices.

a. Write down the name of the devices and the category (input, output, or storage) each falls in.

b. Match these devices with the tasks they can perform.

c. Discuss devices that can be used to perform the same tasks.

ii. Watch the video on how to assemble and disassemble the desktop computer.

iii. In groups, write a set of instructions based on the assembly process you saw in the video.

CATEGORIES OF COMPUTER SOFTWARE

See your computer as a human body. The hardware we discussed earlier is like the body's physical structure – the head, legs, hands, mouth and many more. Software is the soul that gives life to the hardware of the computer system to work. Without the hardware, the software cannot function and vice versa.

In today's lesson, we will discover the different types of software and see how they are seamlessly integrated into various aspects of our lives. From enhancing education and streamlining our work to enriching our personal lives, software is the invisible force that makes everything run smoothly in the computer.

So, get ready to explore the soul of your computer and see how it brings the hardware to life in ways that make our daily lives easier, more efficient, and more connected.

Categories of Computer Software

Now let's take a look at the two main categories of computer software as shown in **Figure 2.39 –** Applications and System.

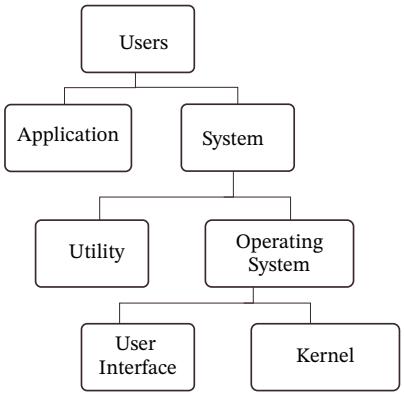


Figure 2.39 Categories of Computer Software.

Application Software

Think of application software like different tools in a toolbox, each tool performing a unique task.

In the same way, **Application software** are programs designed to carry out a specific task other than one relating to the operation of the computer/device itself. Application software is tailored to fulfil particular user needs, such as productivity, communication, entertainment, education, and more. Examples of application software include internet

browser, a spreadsheet program, database management systems, a word processor, and games software. A computer used to manage inventory for a manufacturing company will have different applications compared to one used by an electrical engineer.

Application software runs on top of the systems software, an operating system and utilises system resources to execute tasks according to user instructions.

Here are examples of some categories of application software.

1. Productivity Software

Productivity software helps users create, edit, manage, and share various types of digital content, including documents, spreadsheets, presentations, and databases. Examples include Microsoft Office Suite (Word, Excel, PowerPoint), Google Workspace (Docs, Sheets, Slides), and Adobe Acrobat.

2. Multimedia Software

Multimedia software enables users to create, edit, organise, and playback multimedia content such as audio, video, and images. Examples include media players (VLC Media Player, Windows Media Player), photo editing software (Adobe Photoshop, GIMP), and video editing tools (Adobe Premiere Pro, iMovie).

3. Educational Software:

Educational software is designed to support teaching and learning activities by providing interactive tutorials, simulations, quizzes, and educational games. Examples include learning management systems (Moodle, Canvas), educational apps (Khan Academy, Duolingo), and digital textbooks.

Systems Software

System software serves as the foundational framework of a computer system, much like the foundation of a building upon which bricks are laid. These bricks represent the programs that perform specific tasks within the structured support of the system software. System software ensures computers run smoothly by overseeing hardware functions and serving as a foundation for running additional programs.

Systems software is the program that governs the computer system. Allowing the computer to:

1. control the hardware, including any peripherals
2. allow application software to run
3. provide an interface for the user to interact with the computer
4. maintain the system

In essence, system software acts as an intermediary between the hardware and the end-user applications, enabling the efficient execution of tasks and providing essential services for the computer system to function properly.

Now let's take a look at Examples some of important systems software.

1. Operating Systems (OS)

An Operating System (OS) provides the user interface, manages hardware resources, and manages the running of applications.

In order to perform the actions requested by the computer's users, an operating system must be able to communicate with those users. The portion of an operating system that handles this communication is often called the *user interface*. Older user interfaces, called shells, communicated with users through textual messages using a keyboard and monitor screen.

Nowadays, computer systems usually perform this task by means of a *Graphical User Interface* (GUI—pronounced "GOO–ee"). In GUI systems, applications run in Windows, and all objects (apps, hardware and files) are represented by icons. Application features are accessible through the use of menus. Users interact with the interface by using a mouse and on-screen pointer.

An example of a hardware resource managed by the OS is RAM. When a program is run, it is loaded into RAM. The operating system determines how much memory the program requires and allocates enough space to hold it and its data. When the program is closed, the allocated space is freed up for use by other programs.

Multitasking means being able to run two or more programs simultaneously. An example would be a user streaming music while entering text into a word processor. All modern operating systems have multitasking capabilities. The OS allocates system resources such as CPU time, input/output devices, and computer memory among the programs currently running.

Examples of operating systems include Microsoft Windows, macOS, Linux, Android, and iOS.



Figure 2.40 Some OS examples.

2. Device Drivers

A device driver is a program that controls a specific hardware device attached to a computer. Device drivers control and facilitate communication between hardware devices and the operating system. They relay requests for device access and actions from the operating system and its active applications to their respective hardware

devices. They also deliver outputs or status/messages from the hardware devices to the operating system and thus to applications.

Devices such as keyboards modems, routers, speakers, and printers require device drivers to operate.

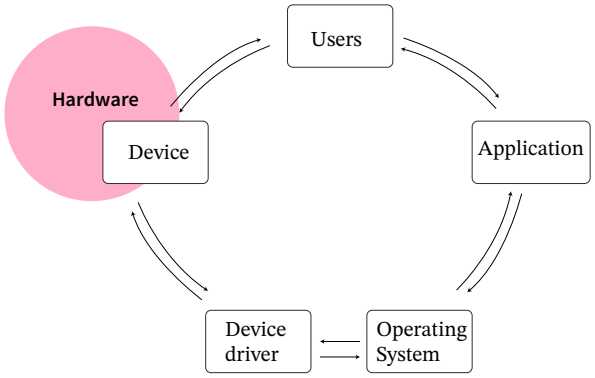


Figure 2.41 A device driver acts as a translator between a hardware device and the applications or operating systems that use it.

3. Utility Software

Utility Software is system software that helps to maintain the proper and smooth functioning of a computer system. These programs assist the operating system to manage, organise, maintain, and optimise the functioning of the computer system.

A file compression program is an example of utility software. Other examples are shown in Figure 2.42.



Figure 2.42 Examples of utility software.

Comparison between utility software and operating systems

a. An **OS is a must-have software to operate a computer, while utility software is optional and can be added as per user convenience.**

b. Utility software assists the operating system but never replaces it.

c. Both are system software, but their functions do not overlap.

Activity 2.13

In pairs, explain the differences between the Android operating system and the WhatsApp application on your mobile phone?

Activity 2.14

Match the following task to their respective software and give 2 examples of software used in each task.

A. playing music
B. word processing
C. protecting computer against viruses and malware.
D. managing the use of the computer's memory.
E. monitors and optimises the performance and health of the hard drive
F. providing a user interface

System software	Application software	Device drivers	Utility

Activity 2.15

Create a presentation explaining how the operating system on your computer or smartphone helps you manage your files and applications?

Activity 2.16

Working with a partner, discuss when was the last time you updated your operating system, and what improvements or changes did you notice? What is the purpose of these updates?

Activity 2.17

Research using the internet to come up with the purpose of an operating system that our mobile phones operate on. Present your findings to the group.